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Libelium Smart World

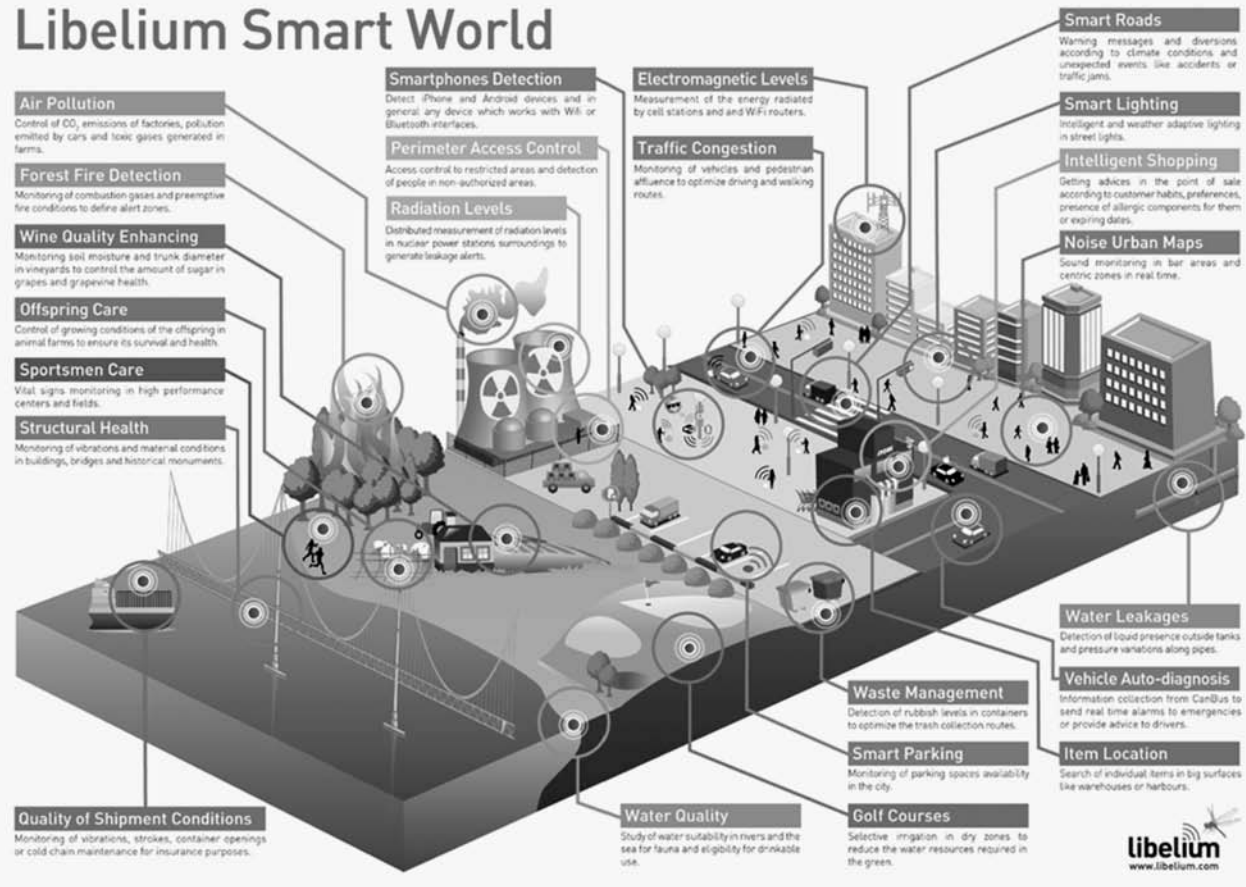


Figure 9.1. Libelium Smart World. Libelium's infographic comprising Smart Cities, Internet of Things, and other sensing applications. http://www.libelium.com/top_50_iiot_sensor_applications_ranking. Copyright Libelium Comunicaciones Distribuidas S.L. 2013.

Digital Infrastructures of Witness

Constructing a Speculative City

PARTICIPATION IN THE DIGITAL CITY, as discussed in the last chapter, is far from a clear and simple project of using digital technologies or platforms for achieving common good. The contributions and pathways of participation unfold along multiple often-circuitous routes, while modes of participation are also entangled with more-than-human entities that may disrupt or prevent engagements. In the emerging smart city—as it is now being implemented—digital sensors are not simply scattered nodes of computational hardware but are also bundled into urban infrastructures in ways that remake these systems and the practices that they animate. The digital infrastructures that variously constitute the smart city, from smart grids to connected devices and smartphones, are also sites where participation is organized and potentially altered and rerouted.

As discussed in chapter 7, many proposals for smart city projects still exist at the speculative or near-future stage. Yet some initiatives have translated into alterations to the built environment, including new services as well as retrofitted urban infrastructures and processes. This chapter looks at sensor technologies and smart city initiatives as they have been implemented and discusses how these distributed modes of environmental sensing influence urban experiences. I specifically discuss sensor projects and smart cities technologies developed in London, a city that is frequently referred to as one of the smartest worldwide, and which has a number of current and ongoing initiatives to implement smart technologies.

I am interested to understand digital infrastructure in these emerging smart city initiatives, not as something that can be defined at the outset, but as something that comes into being through concretizing technological arrangements as well as distinct ways of inhabiting these infrastructures. While it would be possible to make a straightforward list of the typical ways in which this “fourth utility” is unfolding, I suspend these assumptions about what exactly constitutes smart

and digital infrastructure in order to examine more closely how digital infrastructure actually manifests through technologies and practices. Simondon makes the point that an abstract technology is not necessarily a cognitive model that is implemented, but rather is a set of dynamic changes that occur in any given technocultural system to make possible the concretization of particular technologies. In considering the smart city as one such abstract technology, I suggest it is vital to attend to the specific environments in and through which specific smart cities are able to take hold.¹

Furthermore, Simondon suggests that as technologies become concrete they do not become more fixed, but rather become *more indeterminate*.² Abstraction, as it turns out, may constitute a more static rendering of technology than its concrete instantiations, where multiple ways of materializing, practicing, inhabiting, and intersecting with technology can occur. Processes of constructing a speculative smart city constitute ways of working out and transindividuating the entities that populate smart cities and the ways in which they relate and connect up. This chapter turns to consider the transformations that occur as smart cities migrate from an abstract and even speculative set of technologies to more concrete materializations. In relation to implemented sensor technologies, I ask: What are the processes that these infrastructures instigate and sustain? How do they at once individuate and join up cities and citizens? What are the capacities of these infrastructures and what modes of inhabitation do they facilitate?

Part of my objective in asking these questions is to consider the processes whereby digital infrastructures become environmental in the city. This focus takes up the emphasis placed throughout this study on the *environmental* aspects of computation, but here particularly attends to existing urban infrastructures. By “becoming environmental,” I also address much more than a process of spatializing digital technologies, since as I have discussed throughout this study, environments are not merely established backdrops against which the activities of human and more-than-human entities unfold, but rather are involved in distinct processes of becoming and concurring along with entities. Environments are then interconnected with entities inhabiting those environments. Environments are also the conditions in which particular entities may take hold—they ensure the success or the lapsing of particular entities and establish further conditions for invention. It is important to continue to extend the environmental aspects of computation in these ways, since it enables a more dynamic and processual understanding of how environments and digital media concur to form actual entities and actual occasions.

Environments and environmental computation also constitute situations in which “withness” might be articulated and in-form processes of participation. Withness, following Whitehead, is a concept that signals modes of being and becoming together, of concurring, such that the possibilities for both urban

ontological engagements as well as urban speculative futures are undertaken.³ Withness raises the question of how we “possess” the world and become together, not exclusively as a matter of intelligence or rational cogitating actors, but as embodied if differently directed creatures in shared worlds. In this chapter, I attend to the ways in which digital urban infrastructures are productive of occasions of withness. Borrowing this term from Whitehead, I suggest here that withness is a way of moving an analysis of participation to the things, entities, and occasions that are brought together within digital infrastructures and articulating how these modalities of withness influence the types of politics that might in turn unfold. Digital infrastructures in smart city developments induce particular types of participation, which are further productive of modes of withness that generate individuations, embodiments, and inhabitations within the very particular environmental–computational worlds of smart cities.

As discussed in the last chapter, much attention has been directed toward the ways in which digital technologies might activate distinct types of action and engagement, where by virtue of having information and being connected to a seemingly expansive if hazy “community” of fellow users, urban citizens can be better attuned to political projects. But I would suggest that there is much detail that is elided in this characterization of participation, since it assumes a rush to effective action that may never transpire. By attending to the withness of participation, I instead want to identify whom or what particular participatory practices gather together, what these relations or nonrelations might be productive of, and what political contestations are bundled into these groupings. If much participatory literature has attended to the empowerment of individual or collective digital users, this analysis instead considers how a user is only one figure within a wider infrastructural network of participatory and transindividuating politics and action.

While most digital participation focuses on the empowerment of individual users who together make up a collective, or who exercise a self-ness that reinforces notions of doing-it-yourself, here I want to move away *both* from the idea of empowered individuals and from action itself, since the assumption of making citizens and cities productive through optimized actions is so much a part of smart city and digital participation projects that to suspend action in particular is a way not only to ask the Stengerian question in relation to the idiot, “What are we busy doing,”⁴ but also to attend to ways in which urban life might be characterized other than as the ceaseless unfolding of productive (and often economically aligned) individual activity.

Moving from abstract to concrete instantiations of the smart city, I further consider how to move from programs of participation (as discussed in the previous two chapters) to articulations of withness. Participatory, DIY, and smart urbanism projects are articulations that circumscribe withness in particular ways:

they are commitments, with consequences, to ways of being and becoming together. Control systems communicating with transport networks, CCTV performing video analytics for pattern detection, smartphone apps providing access to urban maps and services: in each of these examples, what in the smart city rhetoric would be described through narratives of optimization, facilitation, and efficiency are occasions of withness that express, constitute, and contribute to the ongoing formation of distinct urban individuations and inhabitations.

How are we *with* the (smart) city, its infrastructure, its other inhabitants, and the many computational devices that would steer us, when emphasis is placed on coordinating flows of movement so that stoppage, disruption, breakage, and jamming are minimized? What is the withness of a ceaselessly flowing city, of a city that never stops, that in its automated efficiency continues to process goods, information, and waste in the small hours of the night? Clearly, to discuss digital infrastructures of withness then also requires attending to infrastructure as process. While in some cases emphasis may be drawn toward the physical aspects of infrastructure, with roads, bridges, sewers, electricity, and telecommunications interconnected as basic organizational structures, at the same time a growing body of research now attends to infrastructure as process, which may become more pronounced in moments of failure and breakdown.⁵ Digital infrastructures draw our attention to the ways in which the occasions of cities would not be possible without infrastructural processes. Distinct urban processes are facilitated, enabled, and connected up through these digital infrastructures.

I employ the notion of withness to characterize particular human and more-than-human engagements and concrescences that occur with and through digital infrastructures. I further attend to these infrastructures both as they are emerging as new urban technologies and as they orient urban life toward speculative constructions and additional urban potentialities. In my use of the term “speculation,” as discussed throughout this study, I am particularly concerned to draw out the ways in which the future is already present in current instantiations of smart cities. The future is being made with each sensorized traffic intersection that is implemented, with every smartphone-interacting signage that is developed, and with every prototype and policy document that legislates toward making smart urban environments possible.

Speculation in relation to the smart city is an ongoing practice that, complementing the discussion undertaken in chapter 7 of this study, is both an imaginary and a material-political practice. Speculation is not a project of making fictions but rather is a practice of constructing particular trajectories of urban practice and inhabitation. Construction occurs here in at least two senses: of being built, and of forming the conditions in which new speculative urbanisms (and modes of withness) may unfold. The process of constructing a speculative smart city then points toward experimentation and invention. In the conclusion to this chapter,

I consider how digital infrastructures, smart cities, and participatory urbanisms might be advanced through considering modes of withness that are experimental and that, in a Simondonian sense, might become more *inventive* as indeterminate technical arrangements.

AN INVENTORY OF SENSORS IN THE SMART CITY

Even a notional inventory of sensors in London might reveal that this is a city popping with proverbial intelligence, where the number of sensors in place and in operation suggests that the apparently speculative smart city is already in construction. On any given day traversing the city, one might encounter sensors not just in the smartphones that many people carry (devices that are packed with humidity sensors, temperature sensors, a digital compass, an accelerometer, a gyroscope, GPS, a touch sensor, a microphone, an ambient light sensor, an optical proximity sensor, and an image sensor) but also in every bit of urban infrastructure. Here are multiple amalgamations of sensors and networks, which collectively are meant to add up to more intelligent urbanisms.

An inventory of smart technologies might also incorporate where sensors are located, what functions they enable, how they transform urban processes, what modes of engagement they require or facilitate, who is able to access them and the data they generate, and what happens if they fail or go awry. And indeed, even notionally following such an approach in London, one finds that sensors can be located within preexisting utility infrastructures, including electricity grids and increasing numbers of smart meters; in water mains for leakage and crack detection, as well as flow rates and meters; in waste transfer stations and collection bins; in sewers for tunnel monitoring (and, prospectively, for chemical- and biosensing for drugs, bombs, and diseases); and in an increasing array of smart home technologies that moderate the boundaries between home consumption and utility provision.⁶

Multiple sensors and smart networks are already in use in transit systems in London, from pedestrian crossings for managing the density and flow of foot traffic to bus sensors for alerts on current arrival times (which include GPS location sensors; GPRS transceivers; physical bus measurements via odometer, gyrometer, and turn rate sensors; and mobile IP connections for transmitting data from eight thousand buses to the CentreComm command center).⁷ There are sensors on the cycle-hire docking stations and on buses to avoid collisions with cyclists. Sensors are in regular use at traffic lights and as cameras for live-traffic feeds.⁸ Sensors are further dotted throughout the London Underground on escalators, in tunnels and on lifts, on HVAC control systems, and joined up with loudspeakers and CCTV. In this extended Underground system and sensor network, data on temperature, vibration, humidity, faults, system alerts, and equipment degradation are fed live to a secure cloud for further integration in order to monitor and automate tasks.⁹

Sensors are located on building sites, to monitor vibration and strain from piling, and at tunnel construction sites.¹⁰ There are sensors on parking bays to signal availability in order to discourage drivers from circling to find a parking spot.¹¹ Innumerable CCTV cameras operate as sensors (as discussed in chapter 2), and video analytics are used for crowd management to detect suspicious activity and identify hazards or accidents.¹² Sensors are also in use for multiple forms of monitoring related to health and the environment, including the LAQN air-monitoring network discussed in chapter 6, as well as sensors at Environment Agency sites for monitoring river levels, water quality,¹³ and air pollution in relation to industrial activity.¹⁴ Sensors monitor for vibrations and displacement at the Thames Barrier¹⁵ and the wind speed and direction on the London Eye.¹⁶ Multiple sensors are at work to facilitate weather monitoring, including stations from the Met Office as well as smaller weather stations, amateur stations feeding into Weather Underground, and mobile phones set up as weather stations feeding into crowdsourced maps.¹⁷

Sensors can be found overlapping with existing infrastructures, in some cases forming new networks; sensors are in place to monitor specific temporary uses and events such as construction; sensors are monitoring air and vibration and water levels; and sensors are carried around in smartphones, as wearables, and other portable devices, whether as DIY citizen-sensing tools or monitors for detecting specific phenomena. Sensors are also not the only source for data generated to manage urban systems—alternative data sources include both static and dynamic data collected from social media streams, participatory-sensing systems, and predictive and strategic modeling capabilities.¹⁸

Additional London-based smart city initiatives include Living PlanIT, a project developed out of Formula One racing technology and put to new use at the London City Airport. This project consists of using data, smartphones, apps, and retail schemes to maximize shopping opportunities for High Net Worth individuals, as well as a schematic plan for an “urban operating system” for London, together with energy, water, waste, mobility, security, building controls, analytics, and apps.¹⁹ In a somewhat different arena, the Intel Collaborative Research Institute (ICRI) for the Sustainable Connected Cities project is a collaborative venture across Intel, University College London (UCL), and Imperial that includes projects such as “living labs” for testing air quality in London.²⁰ A video documenting the Intel initiative argues that cities such as London are “reaching capacity,” starting to “break down,” and so need to be “more efficient.” There are even more projects than this underway that demonstrate the extent to which London is an active site for experimenting with, prototyping, implementing, and trialing sensor-based smart cities projects.

These sensor architectures and developments are, of course, situated within a larger context of multiple smart cities pilot projects and plans underway in London, from the Smart London Plan developed by the Greater London Authority to the

Sensing London initiatives advanced by the Future Cities Catapult, funded through the Technology Strategy Board.²¹ The Smart London Plan, for instance, outlines the rapidly growing population of London, which is expected to expand by one million by 2021, as well as numerous other urban constraints that smart infrastructure is meant to help manage, primarily in the form of “mobile internet applications, the internet-of-things, cloud computing and insights from big data.”²²

With this extensive yet by no means complete inventory of sensors as well as other data streams increasingly informing the urban processes of London, it is still not entirely clear at what point this proliferation of technologies might cross a threshold to constitute digital infrastructure or fully formed smart city developments. As mentioned in the introduction to this chapter, a list of sensor technologies does not necessarily point automatically to infrastructural development and may instead raise the question of to what extent sensors must concretize in order to constitute smart infrastructure. Some of these developments are bundled onto existing infrastructure to remake it either as more efficient or more readily maintained. Other developments enable emergent forms of connectivity that form largely invisible networks within everyday operations. But at what point would this array of sensors tip over into smart infrastructure and smart city-ness? Do infrastructures within cities further need to become their own self-regulating organisms in order to qualify as smart?

London is of course not alone in the proliferation of smart cities projects underway, and (as discussed in chapter 7) farther afield there are notable developments in Santander, which has been dubbed the “smartest city” in Europe;²³ in Songdo, Korea, which developed an entire smart urban development from scratch on a green field site;²⁴ in Rio de Janeiro, where the famous “control room” has by now circulated as a pervasive image of smart cities in operation;²⁵ in Dublin, where Intel has created multiple sensor “gateways” and a CityWatch platform;²⁶ and in multiple smaller cities such as Dubuque, Iowa, and Milwaukee, Wisconsin, where IBM has undertaken an approach that consists of meeting with urban planners and community members to identify and document urban issues that might become sites for smart city intervention and technological development.²⁷

While Intel, along with many other advocates of smart cities technologies, have variously suggested that cities are at breaking points and these technologies will facilitate efficiency and added capacity, the cities in which sensor technologies and networks are being implemented represent a wide range of urban circumstances, from the admittedly often-overloaded infrastructures of London to the seaside midsize urbanism of Santander to the relatively low-density urbanisms of midwestern locations such as Dubuque and Milwaukee. These cities are not all alike, but their need for sensor and smart technology has been largely narrated through the pervasive urban problematic of cities at a breaking point due to demographic change, climate change, and rapid urbanization. Perhaps it might



Figure 9.2. Siemens Crystal, a smart and sustainable building in the Royal Docks area of east London. Photograph by author.

be necessary to look more closely at what a concentration of sensors entails within a specific urban site in London.

Meeting Designo at the Crystal

Seeking to gain a more detailed understanding of how these sensor and smart technologies unfold in built environments, I identified one site in London to learn how the specific urban dynamics in this location have coincided with the development and operation of new digital infrastructures. On the north side of the River Thames, in a shiny-spaceship hulk of a building situated in London's Green Enterprise Zone in the Royal Docks, the Siemens Crystal rises as an icon of both sustainable and futuristic urbanism.²⁸ The development consists of the Crystal building, which opened in autumn 2012 as a model of sustainable and smart architecture, as well as an exhibition on smart urbanism that documents smart cities as they are unfolding and might continue to develop in the future.

Like most of London, the Royal Docks is an area of intensive development and ongoing real estate speculation. The Emirates cable cars that pass overhead at this site were developed for the 2012 London Olympics, and new housing and

office developments circle the riverside grounds of the Crystal. Further development plans are proposed for a new Silvertown Tunnel to connect the north and south banks of the Thames, along with an expanded London City Airport nearby and a £1.7 billion Chinese development in this enterprise zone to establish a base for Asian businesses.²⁹ Smartness inevitably becomes bundled into these ongoing plans and proposals for urban development and expansion.

I am visiting the Crystal on this particular day to take a “technical tour” of the smart and sustainable building that is meant to also be a showpiece and the centerpiece of the exhibition on smart and sustainable cities. One could say that before I even arrived at the Crystal my journey crossed through several levels of smart technology, from planning my journey online with the London TfL search tool providing real-time travel conditions, to receiving live arrival times for the Docklands Light Railway (DLR) at the station, to the Oyster card RFID reader used to access transport, the CCTV cameras under whose watchful gaze I fell, and the automated driverless running of the light rail trains that transported me to the Crystal. As I later learn from the technical tour, before I had even entered the Crystal building, I had also already entered the first of four levels of smart security by approaching the “SiteIQ” bubble that surrounded the site, a field of cameras and motion sensors working through algorithms and Video Motion Detection (VMD) to parse whether I might be a suspicious entity, present at the wrong time of day, or incorrectly placed or shaped. If I had set off any of the pattern criteria, cameras would be informed to track me and alert security guards to attend to my movements.³⁰

Having passed through this layer of security, I enter through the automated doors to sign in for my technical tour. In addition to my visitor pass, I am given a smart card for the exhibition so that I may touch in to gain additional information about each display, while leaving a data footprint at every display I visit. The Crystal, I learn from the tour guide, is a building that is meant to “respond to nature,” in its crystalline shape. The structure is supplied with a number of sustainable technologies, from a rooftop that harvests rainwater to be funneled to an underground thirty-thousand-liter rainwater tank to supply water for the building; to an extensive array of solar panels on the rooftop that provides a variable amount of energy, depending upon time of year; to the seventeen kilometers of pipes underground used for the ground-source heat pump and the automated ventilation system that responds to internal and external conditions to cool and heat the building. Sustainability and smartness are once again paired in this digital-urban infrastructure.

All of these technologies are watched over and coordinated by a building management system developed by Siemens named Desigo, which monitors eleven thousand points around the building. A sort of uber-cybernetic control system, Desigo controls the ventilation, lighting, heat, and much more by responding to

sensor data provided by the Building Automation Solutions (BAS) sensors located throughout the building. As part of the technical tour, I visit a conference room to learn about Desigo. The Desigo system is projected from a laptop, since it is accessible from any device connected into the network (and much of the operational oversight of Desigo actually takes place not in London, but in Frankfurt, where Siemens is headquartered). Scrolling through the Desigo system, the tour guide selects a conference room on the second floor to zoom into. We are able to see the trench heating and chilled beams that are the basis for the heating and cooling of the building, as well as the forty-one sensors that monitor this particular bit of infrastructure to ensure stable temperatures.

From the Desigo panel, it is possible to set points to change the temperature across the microclimates of the building, to turn lights on and off, to see if windows are closed or open and to operate them from the building management system, to investigate motion sensors that may have been triggered, to observe activity levels in office areas, and to keep track of a log of failures. Desigo also logs in real time the amount of electricity generated from the photovoltaic rooftop installation, which on this particular day is producing between 3 to 6 percent of the building's electricity. A weather station site indicates the local external weather conditions. The system senses external and internal factors such as CO₂, light

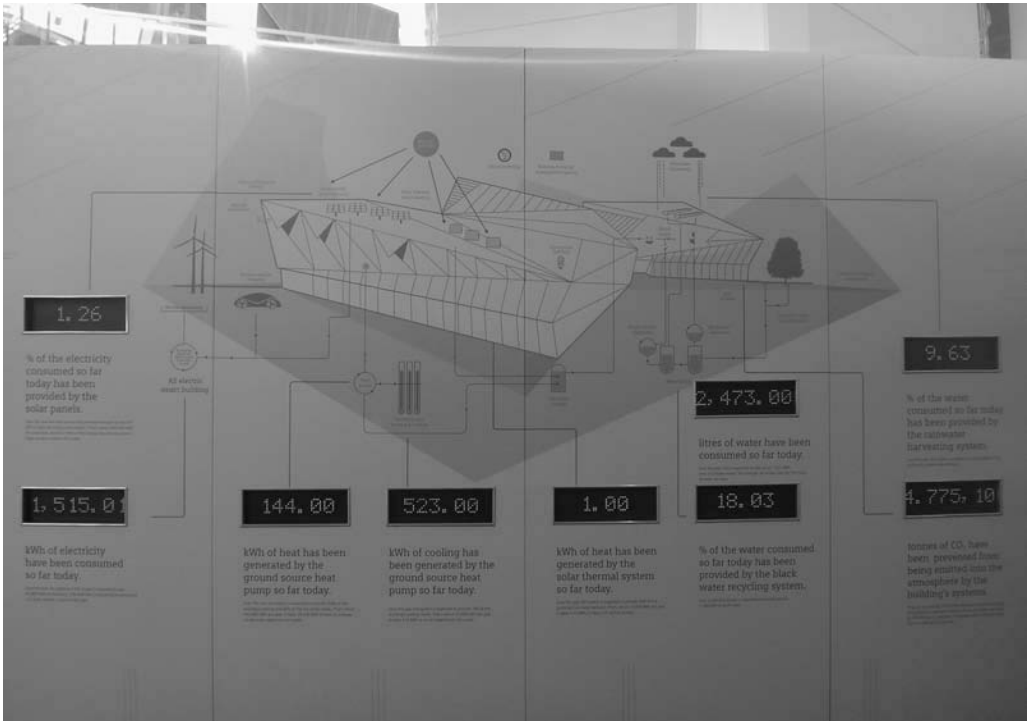


Figure 9.3. Siemens Crystal display of monitored and automated building processes. Photograph by author.

levels, rain intensity and direction, and wind speed and direction, as well as temperature and humidity. Algorithms track sense data to respond to conditions and adjust ventilation in order to maintain comfortable indoor temperatures.

This central automated system is a sort of system of systems, a tool for monitoring and managing the building that largely takes place free from human intervention. While this control system does not need watching over, it can be used remotely—from an app, laptop, or PC—to make adjustments. The building management system largely “takes care of itself,” as it senses, adjusts, manages, and maintains a reasonably homeostatic environment. Emerging with this building are orders of smartness, the intelligence of an individual sensor multiplied through a distributed sensor network; and the intelligence of a sensor network multiplied through an automated building system that is self-regulating. Smartness becomes a matter of intelligent automation, of a cybernetic organism that is able to monitor, manage, and adapt to circumstances in real time. This self-regulatory capacity is not just a management tool and ethos for the building, but also for the city, which as the Siemens exhibition would inform me, could benefit from the self-regulating abilities of sensor networks, where the city itself might become a cybernetic organism.



Figure 9.4. *Play the City Game* exhibit, Siemens Crystal. A game for managing the city according to four key indicators for sustainability, which can be altered via a dashboard. Photograph by author.

Play the City Game

After I have completed the technical tour of the building, I turn to spend some time in the Crystal exhibition, which is advertised in the promotional literature as the “world’s largest exhibition devoted to urban sustainability.”³¹ This is a highly technologized and managed version of sustainability, where technological solutions are presented as the key way to help the city balance considerations for the environment, economy, and quality of life.³² The exhibition begins in the Forces of Change theater, where the “megatrends” of demographic change, climate change, and urbanization are explored as key factors “that determine our future” and are putting pressure on cities, forcing them to adapt in new ways. Here is the recurrent narrative of cities at the breaking point, expressed not just by Siemens, but also by Intel, Cisco, IBM, and a host of other technology companies. Cities are systems, similar to natural ecosystems, the exhibition informs its viewers, and so automated feedback loops are a necessary way to ensure the regulation and management of urban systems.³³

Following on from this first part of the exhibition to the ground floor, the subsequent part of the exhibition focuses on “creating cities,” where importance is placed on understanding the “overall system” in order to manage and address buildings, transport systems, distribution of energy, and water—the vital components in making a city work.³⁴ A central city icon is located in the center of the exhibition space to signal the importance of this overall view for achieving a “well-designed city.” This abstracted urban scene appears in the form of a black tower coursing with an array of colored lights, where the city resembles computer circuitry neatly connected up and pulsing with activity. Here, the city has been remade in electronic form in order to merge with current technological developments.

Planners, the exhibition text and book indicate, are also key to “steering” a city in the right direction.³⁵ In this part of the exhibition it is then possible to play the game of the future city, and to put oneself “in the shoes of a city manager.”³⁶ This hypothetical city scenario has 3.5 million citizens and \$30 billion to administer. In the managing game, the player-planner has forty years and four critical areas to manage: power and water, security, transport, and finance. By adjusting these levers on a dashboard, a status menu indicates how well one is doing with this city management exercise and in addressing urban problems overall. Notably, every urban resource, infrastructure, and problem that is to be addressed must be quantified in order to be made computable. As the booklet accompanying the exhibition notes (citing ex-mayor Bloomberg), “If you can’t measure it you can’t manage it.”³⁷ Urban life must be enumerated in order to be managed within this cybernetic system. But once measured, the city is meant to emerge as an easily pliable and modifiable system.

Beyond this game of running your own city, there are displays on smart buildings—since buildings consume 40 percent of the world’s energy—and the Crystal is an example of how buildings might become more sustainable. There are additional displays on security and crowd control, electricity and the necessity of reconfiguring energy infrastructure to a smart grid, on water and reducing water consumption, on aging and the development of scanning lab tests and genetic profiling to attempt to reduce strains on health care, on measures to improve air quality and reduce climate change—inducing emissions, and on urban transport.

At the end of the exhibition is a Future Life gallery, where New York, London, and Copenhagen are presented circa 2050 as cities where many advanced technologies have been implemented in order to achieve greater sustainability. Future urban scenarios include vertical cities-within-cities that have sprung up to accommodate burgeoning populations, smart grids that respond to fluctuations in renewable energy supply by alerting residents to use energy during a surplus or conserve at peak times, and sensors that make for an aware city—“a seeing, hearing, thinking, feeling, living organism” that is able to respond intelligently to real-time information as it “flows into the city cockpit.” All aspects of urban life have been made sustainable, from parks to energy, transport, waterways, and even citizen participation, where “the city responds to the needs of the people.” The city also never sleeps, as it efficiently and automatically activates, restocks, recharges, and recycles during the night.

The smart city as built and imagined seems to toggle in this in-between zone, instantiated in some ways but always leaning toward a more complete automation, a more fully self-regulating (and so sustainable) organism that monitors, responds, and adapts in real time in order to achieve the most efficient and optimized balance of resources, time, and money. Sensors becoming networks becoming smart cities all appear to be on a trajectory toward an urban organism that acquires an uncanny intelligence and ability to manage the city as planner, architect, and engineer all rolled into one. But when might this phase-change occur, when sensors and networks take on a life of their own and begin to organize their own automated processes within the city? Smart cities projects seem to suggest that we are all waiting for the moment when the city becomes the ultimate automated organism, where, as Norbert Wiener would suggest, all the sense organs that might constitute individual and assorted automatons participate in and feedback into a larger cybernetic system.³⁸ Digital infrastructures are part of this accumulation of automaticity, where new structures, organizations, and processes of digital connectivity come together to articulate particular modes of *withness* for subjects (human and nonhuman) and to advance particular ways of “possessing”—and making—worlds.

Leaving the Crystal and its current and future vision of the automated and smart city, I take the Emirates cable car across the river to North Greenwich.

Passing over the waste transfer stations and scrap yards, the bits of real estate in between an industrial past and a still-to-be-fully-instantiated future, I sense a deconcentration of smartness as I surface from the Underground en route to New Cross in South London. Here, far from the sustainable smartness of the Crystal, the connectedness, flow, and seamlessness of the city unravels a bit, where streets are clogged with traffic, housing stock is crumbling, and air pollution is a recurring problem. Among the dilapidated and disused phone booths, convenience shops, and ancient laundromats, I sense that the pacing of this area is less geared toward moving bodies according to some well-tuned and optimized circuitry, since the economies here are of a different and less-privileged sort. How would an automated urban organism deal with this indelible aspect of urban life—that of difference and inequality?

DIGITAL INFRASTRUCTURES OF WITHNESS

As mentioned in the introduction to this chapter, infrastructure is commonly described through resources and what it moves or structures, from energy to water, waste, transport, and communications. When digital infrastructure is described as the “fourth utility,” however, it is clear the implementation of urban digital technologies is more than just another piping and cabling job. In this next section, I would like to return to the idea of how an abstract technology or technologies become concrete, and how this process increases indeterminacy. Part of this indeterminacy may actually be understood through the transformations that take place as digital infrastructures take hold and in the numerous entities and relations that proliferate as transindividuations of smart city technologies. These ways of articulating and connecting up entities are constitutive of modalities of withness: of relating, being, and becoming together in the smart city. I now turn to discuss the concretization of digital infrastructures through three modalities of withness—measurement, automatism, and contingency—in order to consider the distinct traversals that digital infrastructures make as they becomes more present in the city. This is a way of saying that we could consider the infrastructural aspects of smart cities not simply as sensors and networks but perhaps also as processes of measurement, automatism, and contingency, which are operations that generate distinct modes of withness. Further to this point, digital infrastructure does not become a matter of hardware and software, physical structure and code, monument and process, but rather of transductive articulations of urban environments, technologies, and inhabitants.

Infrastructure as Measurement

From the TfL network to the Desigo building management system, the sensors proliferating throughout London are assembling into a series of infrastructures engaged in measurement. The rate of vibration, light levels, temperatures inside

and outside, air quality levels, and more: the city becomes a set of enumerated data and variables to monitor and manage in real time, as well as to anticipate through predictive modeling. What will traffic density levels likely consist of in two weeks? How will increased temperatures affect air quality? How will energy levels shift with changing temperatures?

While the focus with machine-to-machine communication has largely emphasized the machines or things that are talking to each other, in many ways there are a series of emerging networks and infrastructures that are forming in and through the drive to measure the city in order to manage it. Infrastructure has arguably never been without its modes of calculation, but with digital infrastructures measurement becomes even more infrastructural—it is a condition and resource that enables social processes and organization. Infrastructure in this sense is transversal, characterized not just by a particular resource to be moved and apportioned, whether water or energy, but also by the overlapping, intersecting, and (ideally) interoperable streams of data that allow correlations to be made across previously separate infrastructural milieus.

Or at least, this is the promise. To date, with smart cities as well as the Internet of Things, some commentators note that it is not always clear what is being measured, what should even be taken into account, and how data might be made interoperable.³⁹ While digital capacities of enumeration folded into infrastructure are meant to be a way of managing urban overload as well as the decay of urban systems,⁴⁰ the sensor networks and accompanying data are not yet at the point of delivering on this expectation.⁴¹ Rob van Kranenburg has suggested that “Ambient Intelligence” and the Internet of Things present the problem of deciding which “connectivities we really want as human beings on this planet.”⁴² Measurement is a primary way in which machine-to-machine and ambient connectivity are unfolding in the smart city and via digital infrastructures. But what does this mode of connectivity involve, exactly? And how does measurement constitute experience (as discussed in chapter 4)?

The measurements that sensors and networks undertake could be described as a process of “taking into account,” following Whitehead (and Stengers), where environments are processed through a subject-superject relation. On one level it might be possible to suggest that each individual device is “taking account” by measuring environmental variables, which eventually add up to “big data.” If we take seriously Whitehead’s invitation to extend subjectness to all entities and not just to humans, then in the smart city we would consider how a location sensor takes account of a moving vehicle, how a light sensor takes account of the sun coming into the windows, how a vibration sensor takes account of the movement of subterranean tunnels, how a sensor network takes account of the multiple streams of data, how an algorithm takes account of distinct patterns in the data, how a program takes account of the conditions of responsiveness in order to

implement, actuate, and thereby continue to change and influence the urban environment that is monitored and managed.

But measurement might also signal the ways in which a collective urban-environmental potential has been parsed to be made into data—itemized, quantified, networked, and operationalized—so that individuals and relations are formed through a collective made measurable. This point draws on Simondon’s discussion of how relations are not formed through the adding up of individuals to form collectives. Rather, collectives are transindividuated into distinct entities, and it is this mode of parsing collective potential that in-forms individuals and relations. So rather than decide which connectivities are preferred between already enumerated individuals, we might instead attend to the ways in which collectives are turned into measurable entities and individuals, which are further put into relation through infrastructures of measurement.

Infrastructure as Automatism

Infrastructures of measurement are not simply expressions of counting or taking into account, moreover, but are also about engaging in processes whereby that which is enumerated is also capable of becoming *automated*. One of the primary commitments of smart city developments is to automate urban processes so that responses to conditions of overload, situations requiring rerouting, or moments of alarm can be reacted to automatically, through sensors that detect and actuators that respond. Automatism in this way is a key aspect of how Wiener developed his ideas of cybernetics, where “servo-mechanisms” might perform automatic functions in relation to identified triggers. Within these feedback loops, Wiener also integrated humans as sensor-actuators as part of a cybernetic system. The reactions of pilots in warfare, for instance, became one of his areas of focus, where automatism could be extended to a body-machine loop of triggers and actions.⁴³

The logic of automatism then points to the ways in which regulation might be achieved, so that urban-infrastructureal-human arrangements might unfold more seamlessly, and disruptions might be minimized. In this way, urban systems are managed as cybernetic entities. The smart city implements numerous instances of sensors and humans actuating both other machinic functions, as well as further human actions. Wiener’s cybernetic pilot functions through an integrated relationship with a dashboard, where responses emerge in relation to immediate triggers. Cybernetic dashboards are also proliferating in smart cities, where not just the famous control room of Rio but also the more platform-based dashboards that consolidate multiple urban sensor streams become sites for inputting and outputting, for gathering sensor streams in order to respond or actuate in relation to sensor variables. Smart buildings such as the Siemens Crystal has its Desigo control system and dashboard, and London even has several dashboards, including

the City Dashboard project, where data from weather stations, tube status, London cycle hire, DEFRA air pollution stations, river levels, Yahoo stock data, traffic cameras, Twitter trends, BBC News, Open Street Map, the electricity grid demand, and even a mood index indicating happiness levels, are amalgamated into both a dashboard and map view.⁴⁴ The data made available here are collected from sensor and web feeds, which provide an apparent overview of the city while at the same time indicating a whole range of automated urban processes underway or that might be further managed.

The dashboard-as-platform also demonstrates how the city is becoming a platform. While Internet platforms for social interaction are one aspect of smart city initiatives, platforms also unfold and are distributed across urban space. The city as platform has been a topic of discussion in recent seminars and events sponsored by Microsoft, IBM, and others. Platforms, in these scenarios, are not just Internet-spaces but are also embedded, situational, context-focused applications that map new digital functionalities onto urban infrastructures, processes, and exchanges. Platforms are both localized and distributed in the city and across the Internet. As another layer of infrastructure that enhances the efficiency and timing of cities, digital connectivity and platforms present the possibility of a well-regulated city that becomes sustainable through the enhanced synchronicity and expediency of urban systems. New platforms and connectivities arise to facilitate citizen involvement and monitoring of these processes. Platforms further function as monitoring devices, to be used together with everyday urban practices.

Automatism cuts across these different types and distributions of infrastructure, and a city dashboard or platform signals the many automated urban functions underway. Capturing the potential and processual aspects of infrastructure, the architect Keller Easterling has suggested, "Designing infrastructure is designing action."⁴⁵ This statement could easily be read in a deterministic way, where the structures of infrastructure are seen to offer automatic scripts or codes for action. But the statement could also be read less causally and more simultaneously, where infrastructures and actions coincide as entangled and co-emergent processes. A study of infrastructure could very well attend to the actions that are productive of infrastructures, as well as infrastructures that are productive of actions. This raises the question of how actions unfold within and through automated urban infrastructures. Various smart city developments put forward scenarios where the movements, timings, and circulations of bodies and systems will sync up more easily, where crowd control or shopping may be streamlined through apps and code and CCTV, where the city will operate as a real-time organism, where the witness of bodies in cities will be advanced through the seamlessness that automatism enables.

Simondon suggests, however, that automatism is actually quite a low-level and limited way of engaging with and thinking about technology. As someone

relatively critical of cybernetics and the master-slave relationship it tends to propagate in relation to technology, Simondon suggests that technology is far more inventive when it is open and indeterminate. Limiting technology to “utensil” status is a way to ensure that the technical is also separated from cultural concerns, which might otherwise ensure a wider set of engagements with technology.⁴⁶ As Simondon writes, “In order to make a machine automatic, it is necessary to sacrifice many of its functional possibilities and many of its possible uses.” Instead, he writes:

the real perfecting of machines, which we can say raises the level of technicality, has nothing to do with an increase in automatism but, on the contrary, relates to the fact that the functioning of the machine conceals a certain margin of indetermination. It is such a margin that allows for the machine’s sensitivity to outside information. It is this sensitivity to information on the part of machines, much more than any increase in automatism that makes possible a technical ensemble. A purely automatic machine completely closed in on itself in a predetermined operation could only give summary results. The machine with superior technicality is an open machine, and the ensemble of open machines assumes man as permanent organizer and as a living interpreter of the inter-relationships of machines.⁴⁷

Simondon imagines a more engaging set of operations for machines that requires openness, and in particular an openness to human involvement. One could argue that with machine-learning, the participation of humans with machines need not be described in this way—that machines could be open to the unfolding processes of other machines. Or one could argue that Simondon brings us back to a romantic version of humanism, where humans in the loop present the ways in which machinic closure may be overcome. But as Simondon’s writing indicates through his discussion of the preindividual reserve, what counts as “human” is also not fixed or settled, since machinic engagements also give rise to distinct transindividuations of the entities involved.

In a rather different vein, Wiener develops a trajectory of different versions of automata, from clocks to opening doors, and from photocells to computers, which as servo-mechanisms have operated as sense organs that, when coupled to the outside world, translate information into a series of actions.⁴⁸ For Wiener, automation occurs across the sensor-actuator exchange. Yet as Simondon’s approach to technology suggests, this relationship of automatism would necessarily have to be exploded—and perhaps even von Neumann’s computer architecture diagram would need to be redrawn—in order to gather the “responsible and inventive” input of (transindividuating) humans into these circuits.⁴⁹ Technicity—and by extension, computation—is not always a closed loop of action and reaction,

and may just as often open into new orders of indeterminacy, especially as it becomes environmental.

A smart city is meant to join up structures, processes, mechanisms, and gadgets, so that they function more efficiently as some well-timed machine. Infrastructures emerge as a set of relations, but these are not necessarily primarily human relations. Humans are typically flattened into this relationality, not as exceptional actors or beings, but as one more parcel to move along, one more node of information, one more sensor-actuator operation taking place in the city. In this galaxy of machines talking to machines, contra the city game and steering planner of the Crystal exhibition, it is not a human that would cybernetically govern the smart city, but rather an automated system of control where a human operator is one source of input, since many of the protocols and terms of relation are written through the ways in which machines talk (and must talk to) to other machines in order to be interoperable.

Measurement and automatism as modes of connectivity signal the ways in which infrastructures are then necessarily discussed as relations. Writing on infrastructure and the relations it sets in play, urban sociologist AbdouMaliq Simone suggests that the characterization of infrastructure as “in-between” is provocative for considering how cities come together and hold together.⁵⁰ We might also read Simone through another Simondonian register, to consider how this in-betweenness might be understood less as the glue between individuals and more as a parsing of collective potential. In-betweenness is about the relationality that occurs through infrastructure. As Simondon has suggested, not only does relationality not precede the act of relating but the terms or individuals related are similarly not fixed in advance but come about through transindividuating processes that give rise to singular and collective human and nonhuman individuals, as well as the conditions of relationality. With this inverted sense of how relation might be seen to materialize, what other sorts of relations might concreate through the digital infrastructure of sensorized and smart cities?

Infrastructure as Contingency

Simondon suggests that as technology moves from abstraction to concretization it becomes more indeterminate and that this process is not linear. As technology concretizes, relations are differently manifested and materialized in ways that might be described as contingent: contingent upon the environments, entities, and technologies that transindividuate together. For instance, contra those studies that would see code as a discursive program to be implemented in urban sites, a Simondonian approach to concretization would instead ask how code transindividuates along with the entities, environments, and relations in and through which code is meant to take hold. Concretization is not the rational implementation of a plan of technology but rather an actual occasion of witness, a particular

possession of the world that is less scripted and more generative—potentially even of new types of indeterminacy.

Contingency then is a key way in which the concretization of digital infrastructure can be understood. Infrastructural arrangements might be differently constituted depending upon how sensors concreate into organizing structures. Along with infrastructure, a corpuscular society of sensing entities also concretes. Infrastructure is a condition that enables inhabitations, modes of witness, and worlds to be sustained as modes of being and becoming. But encountering infrastructure as concretization and concrecence is not, arguably, synonymous with infrastructure as assemblage, as previously discussed throughout *Program Earth*. Whereas assemblages might emphasize the adding up of entities, a list of everything that comes together to make infrastructure materialize and operate, a concretization and concrecence of infrastructure places the emphasis differently on the processes and individuations that parse and connect up entities and environments in particular ways, such that the capacities of entities might not even be known in advance of their connecting and relating. Such an approach inevitably emphasizes the contingency and indeterminacy that characterizes such technologies as digital or sensor-based infrastructure.

Perhaps illustrating this point with the most sustained clarity and richness of example is the work of Simone, who focuses on (other-than-smart) cities in the Global South, including Jakarta and Mogadishu. Simone suggests another way of encountering infrastructure as it emerges and is articulated in more contingent and itinerant ways—not as fixed structures, but as provisional, always in process and participatory relations. Simone extends infrastructure to include “people’s activities in the city” since, as he writes, “African cities are characterized by incessantly flexible, mobile, and provisional intersections of residents that operate without clearly delineated notions of how the city is to be inhabited and used.”⁵¹ By extending infrastructure and its provisionality to people, and people living in particular African cities, Simone draws attention to the distinct if changeable and contingent infrastructures that come together in particular places.

The changeability of infrastructure is something that Mackenzie also captures in his discussion of wireless technology by attending to how “wirelessness” changes with each instantiation of wireless technology.⁵² At the same time, the “devices that comprise infrastructure keep changing.”⁵³ Rather than infrastructure necessarily constituting a seamless set of connections, Mackenzie suggests that a whole set of lived and felt transitions take place across infrastructure that are different from the assumed fixed experiences of infrastructure. Instead, wireless infrastructure can be piecemeal, in process, and in need of constant upkeep and repair. While there may be a constant hope for infrastructures that are self-administering, the reality is that maintenance is a continual condition through which infrastructure

is encountered and experienced.⁵⁴ Or, as Graham writes, infrastructure is less stable and more made up of “leaky, partial and heterogeneous entities.”⁵⁵

However, in the context of smart cities, it is not uncommon for technical arrangements to be presented as managing diverse urban circumstances toward seamless sets of relations that would be relatively complete and universal across urban contexts. Smart cities developments assume the same modes of withness, where citizens, cities, and technologies, despite their diversity, regularly intersect through a continuous program of sensor-activated cities. Yet urban life is no doubt not the ceaseless undertaking of universal sensor-actuator tasks to be completed, with Tetris-like packages to be dropped in Tetris-like spaces. In the computational architecture of command-and-control automatism writ large over urban environments, there are as many accidents, disruptions, and breakages as there are seamless connections made. People continue to wander and not simply move from home to office to store and back again. Not all urban citizens are hyper-productive economic subjects. Not all urban spaces operate as one more bit of computer circuitry.

If we were to return to Simondon’s notion of how the concretization of technology may give rise to indetermination, we could then say that the smart city in some ways commits the error of making the abstraction of infrastructure real without accident or indetermination. The universal visions of smart cities typically assume infrastructures are always the same in their striving for optimization. Even imaginaries for participatory digital urbanism, as discussed in chapter 8, do not typically allow for indeterminacy. These schemes might admit some contingencies in the form of local circumstance, but they move toward the same end point of managing and regulating cities in order to achieve efficiencies and solutions. Yet as Simone reminds us in his writings on the practices of infrastructure, these are characterized by situated contingencies, where people may even tinker with and alter the city and its infrastructure.⁵⁶ Contingencies in the smart city may emerge across human and more-than-human registers, moreover, since as the city “plays itself” it no doubt is not simply adapting for optimization but is also generating particular materializations of sensor-spaces, transforming environments through programs of more-than-automatism, and giving rise to proliferating bugs and blockages that are sites for ongoing repair.

CONSTRUCTING A SPECULATIVE CITY

The processes whereby digital technologies become environmental are, on one level, a matter of how technologies are distributed in and among surroundings. But, on another level, and following Simondon, this process does not involve a static arrangement of machines but rather it in-forms the very environments in and through which technologies are distributed such that new environments are

made. Weiser's early discussion of ubiquitous computing proposes not just that an array of "tabs" and "badges" and laptops would proliferate but also that these technologies would "disappear" and "weave themselves into the fabric of everyday life until they are indistinguishable from it."⁵⁷ The ways in which ubiquitous computing would become environmental here consist of integrating with surroundings such that technologies do not require active attention. Rather than seeking to make computing a willful cognitive engagement, Weiser found the success of these computing devices to be exactly their imperceptible integration into experience, where they became infrastructural.⁵⁸

We could say that these infrastructural sensor technologies effectively become the environment with which we would interact, along with the subjects, objects, and the milieus they constitute. As discussed in the introduction to this study, Weiser sought to push computing beyond its presence as a self-contained box—where it merely constituted its own self-enclosed world—and in the process to find ways for it to enhance the world already in existence.⁵⁹ But as Weiser's own text intimates, ubiquitous computing would not simply consist of a set of enhancements, but rather would transindividuate worlds and environments in new ways, thereby creating other versions of the real. As one example of ubiquitous computing elaborated by Weiser indicates:

In our experimental embodied virtuality, doors open only to the right badge wearer, rooms greet people by name, telephone calls can be automatically forwarded to wherever the recipient may be, receptionists actually know where people are, computer terminals retrieve the preferences of whoever is sitting at them, and appointment diaries write themselves.⁶⁰

Here is a concatenation of actions that would arrive at more seamless actions in the workplace. It reads as a list of new modalities of ubiquitous computing and bears an uncanny resemblance to the Internet of Things scenarios developed in smart cities proposals and applications. Yet as Weiser later indicates, "Neither an explication of the principles of ubiquitous computing nor a list of the technologies involved really gives a sense of what it would be like to live in a world full of invisible widgets."⁶¹ New worlds come into being—digital technologies become environmental, and in turn, new environments emerge as milieus distinctly conditioned and transformed through these devices.

Processes whereby technology become environmental and give rise to distinct milieus are then less mediatory and more in-forming and ontogenetic. As mentioned in the beginning of this chapter, making a list of sensor-based digital infrastructure does not necessarily address the distinct ways in which these technologies concretize in and with new environments and environmental conditions. As Weiser's above example of the well-running office indicates, the constitution

of environments also involves the constitution of the field of subject-action-event. Digital technologies play a role in making these relations coalesce and concreate in particular ways. Here, digital infrastructures produce medial relationships that are world-making and world-sustaining, rather than mediating across preestablished entities. In these transindividuating infrastructural relations, the conditions and potential for collectives to form and interact arise.⁶²

I have been drawing together Simondon's discussion of concretization alongside Whitehead's notion of concrescence in order to return to a consideration of the particular "techno-geographical"⁶³ milieus that come together along with smart cities technologies. These concretizations and concrescences are also a way of discussing withness, of thinking about how entities become together. In his study of wireless technology, Mackenzie mobilizes a discussion of conjunctive relations to consider "with" as one type of the many ways in which relations unfold. Drawing on James and Nancy, Mackenzie notes that extending the scope of conjunctive relations is a way to think beyond or against merely conceiving of technology through utility or means.⁶⁴ It is interesting to note that, for James, "withness" is a lower-level relation, where "the lowest grade of universe would be a world of mere *withness*, of which the parts were only strung together by the conjunction 'and.'"⁶⁵ Yet for Whitehead, withness becomes a way to discuss the ways in which worlds are possessed.⁶⁶ If we were to take up Simondon on this point, we would also be obliged to note that relations are not able to precede the moments and entities of relating. Withness even constitutes the subjects that would be resident-actors in these worlds.

In this sense, I mobilize withness as a particular way of bracketing off assumptions about what it means for participation to unfold in the smart city. I see withness more as an articulation of processes of participation that involve becoming together, across an extended array of entities, and setting in motion the connections and inheritances that take hold to become something like urban infrastructure, whether solidified as communication conduits or distributed as seemingly immaterial exchanges of practices. In order for infrastructure to exist and persist, it must draw entities into active participation with it. The city is a complex corpuscular society of inheritances that in-form bodies of sense. Withness captures the lived embodiments of participatory experience, while also going further to indicate the ways in which worlds are made and sustained through particular ways of "possessing" the world. In other words, withness describes ways of making worlds.⁶⁷ Digital infrastructure is a multiple and more-than-human event, which may be further described as a process whereby multiple actual entities become tuned in to shared registers of withness, along with shared practices for making worlds. In fact, this is how Whitehead describes a community, where multiple entities are effectively resonating within and experiencing a shared register of world-making.⁶⁸

But this is by no means a way in which to render neutral the worlds that are formed. Whitehead for his part recognized the power and force, in the form of persuasion that enabled certain worlds to take hold and not others.⁶⁹ Stengers takes up this discussion of power and persuasion within Whitehead to draw parallels to Foucault's discussion of power. Here, it is useful to revisit the analyses made in chapter 7 to bring together both a Foucauldian and Whiteheadian analysis of speculation and power vis-à-vis analyses of infection and persuasion drawn out by Stengers in *Thinking with Whitehead*.⁷⁰ As Foucault has noted, the point in discussing power is less to make an effort to outpace or escape its reach but rather to understand its distributions in order to attend to particular formations it enables—or disallows.⁷¹ Stengers takes up Foucault's discussion of power in relation to Whitehead to consider how not just persuasion but also infection and the gaining of a foothold indicate how power is distributed in and through environments and how environments and entities (including facts) are able to persist.

Participation as a concept and practice frequently involves discussions if not disagreements about forms of *agency*: about people becoming more empowered by 2.0 social media or being duped into contributing free labor to exploitative digital economies, about the more-than-human actors and devices that are also participating along with people in digital media exchanges, and about the ways in which agencies of these devices shape the contours of participation. Rather than focus on agency as a key modality of participation, however, I have redirected this consideration of participatory urbanism toward different registers of withness, in the form of persuasion and infection. I have drawn on and adapted these terms from Whitehead and Stengers, who attend to the ways in which entities con-crece, a process that arguably has less to do with agency *per se* and more to do with ways in which entities become together and are drawn into processes of mutual—if differing and differently manifested—influence. Influence—persuasion and infection—are decidedly different from agency because they do not involve a subject/object *acting on* another—willfully or otherwise—but rather have more to do with transferences (and Simondonian transductions) that occur across not always clearly delineated cause-and-effect or subject-and-object sites or encounters. This means that rather than search between determinist or constructivist approaches to technology, we might attend to how new entities, relations, and modalities of withness contingent on infection and persuasion generate new technological practices, inhabitations, and ways of life.

Smart cities developments such as those underway and implemented in London raise the question of how modalities of withness are realized, and how different registers of withness concretize along with distinct approaches to “smart” technology. “Our” urban future is differently distributed depending upon how close to the machine “we” are.⁷² Those that can speak to it, in its language, stand

a better chance of counting and being taken into account as a relevant node in its networks. Those who do not may find they cannot get a foothold in the world the smart city has made and possessed. Possessing a world is not dissimilar to individuating environments and entities. Rather than adding up relations or making lists of technologies, possessions and transindividuations signal processes of taking into account, distributing power, exercising persuasion, in-forming experience, participating, taking hold, and so securing, even if momentarily, an inhabitation, an environment, a mode of becoming.

Withness has been the speculative inhabitation that I have worked with in this chapter, while examining how modes of becoming together are articulated within smart city developments. The lived practices and politics of smart cities, as they are and as they might be, and when addressed through withness, recast questions about what constitutes the environment of the smart city and rework the practices and entities of participation that unfold there.

Withness also points to the ways in which speculative cities are being constructed continually. As Stengers suggests, speculation as a philosophical proposition requires a “leap of imagination”⁷³ that is not as much a matter of projecting abstract ideas of future possibilities as it is a question of how to precipitate and make possible distinct ways of life. In this sense, speculation becomes entangled with persuasion and power, generating a politics of speculation that manifests in particular ways in smart cities projects. It may be that speculation is in fact made possible through distinct modalities of withness—this is not something undertaken alone—and so we might be particularly attentive to the modes of withness that are propagated and sustained in smart cities and sensor-based urbanisms. This might involve querying the programs of withness and participation that digital infrastructures help to sustain.

This chapter ends by suggesting that withness may be generative of experiments. Withness asks how are we thinking with, being with, and becoming with the smart city. It strikes me that many smart city and participatory urbanism projects are missing exactly this more experimental approach to speculation. Rather than open technology to a multiple array of inhabitations, encounters, and modes of withness, these projects most often reduce technology to a “utensil,” as Simon-don has termed it, or as a project of utility and optimization, rather than of equality, actual quality of life, or even wonder. The conclusion takes up some of these provocations for how to experiment with and speculate toward more inventive environmental inhabitations—computational and otherwise.